Simultaneously Large Greatest Prime Factors in Arbitrarily Long Blocks

A rigorous consequence for OEIS A006530

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Abstract

For any prescribed threshold B and block length L, we prove that the greatest-prime-factor function exceeds B at every position of infinitely many blocks of L consecutive positive integers. In fact, the starting positions contain a complete arithmetic progression and hence have positive lower asymptotic density. This is an elementary application of the Chinese remainder theorem, not a solution of the separate iteration conjecture recorded in OEIS A006530.

Definitions and statement

Let $P^+(n)$ be the largest prime factor of $n \geq 2$, and put $P^+(1) = 1$ as in OEIS A006530. For a set $S \subseteq \mathbb{N}$, write $\underline{d}(S) = \liminf_{X\to\infty} X^{-1}\#(S \cap [1, X])$.

Theorem 1. *For all real $B \geq 1$ and integers $L \geq 1$, the set*

$$S_{B,L} = \{n \geq 1 : P^+(n + j) > B \text{ for every } 0 \leq j < L\}$$

contains a full arithmetic progression of positive integers. In particular $\underline{d}(S_{B,L}) > 0$, and there are infinitely many such blocks.

Proof. Choose L pairwise distinct primes $q_0, \ldots, q_{L-1}$, each strictly larger than B. This is possible by Euclid's theorem on the infinitude of primes. The Chinese remainder theorem gives an integer r with

$$r \equiv -j \pmod{q_j} \qquad (0 \leq j < L).$$

Set $Q = \prod_{j=0}^{L-1} q_j$. Every positive integer $n \equiv r \pmod{Q}$ satisfies $q_j \mid n + j$ for all j. Since $q_j > B \geq 1$, we have $n + j \geq q_j \geq 2$ and

$$P^+(n + j) \geq q_j > B \qquad (0 \leq j < L).$$

Thus the entire positive residue class $r \pmod{Q}$ lies in $S_{B,L}$. Its asymptotic density is $1/Q$, so $\underline{d}(S_{B,L}) \geq 1/Q > 0$. □

Remark 1. *The theorem imposes only a lower bound on the* largest *prime factor. It does not say that the numbers are prime or semiprime, and does not establish any open conjecture concerning iterations of the greatest-prime-factor function. The result is a standard CRT consequence and no novelty or priority is claimed.*

References and verification scope

1. OEIS Foundation, *A006530: Gpf(n): greatest prime dividing n*, `https://oeis.org/A006530` (consulted November 12, 2026).

2. Chinese remainder theorem and Euclid's infinitude of primes: classical standard results.

AI disclosure. This document was generated and internally checked by ChatGPT (OpenAI), without independent human peer review. The proof is elementary and self-contained apart from the stated classical theorems.